Did the Role Fire? Auditing the Manipulation Checks Behind Role-Prompted Multi-Agent Systems

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Abstract

Multi-agent LLM systems give agents cognitive roles, such as critic, verifier, or explorer, and benchmark gains are often read as evidence that the roles did their jobs. We borrow manipulation checks from experimental psychology: an executable, preregistered test of whether a role prompt moves the quantity its operation implies, relative to a length-matched neutral prompt. Across six roles and two model families, one role passes on the primary model and two on the second. An exploratory audit of the checks is more informative than their verdicts. Discriminant validity holds in 1 of 12 role–model cells; three of the six checks reduce to a count of the claims an agent writes; no guard against degenerate passing survived; and the study's most striking reversal is an artifact of a mismatch between prompt and check. A manipulation check should be validated as an instrument before its verdicts are read; we describe preregistered controls for doing so.

Keywords: Multi-agent systems; Role prompting; Manipulation checks; Construct validity

1 The inferential gap

Role-based multi-agent designs assign agents distinct jobs, such as debaters arguing opposite sides or critics reviewing answers (Chan et al., 2024; Liang et al., 2024; Du et al., 2024), and credit benchmark gains to those jobs. A benchmark delta does not show that a role did its job: prompt length, tone, or abstention rates can move a score on their own. Personas often fail to improve performance (Zheng et al., 2024), and disobeying the role specification is a documented multi-agent failure mode (Cemri et al., 2025). A preprint measures per-role fidelity in debate without pre-declared tests or length-matched controls (Dietrich, 2026). Experimental psychology addresses the same problem with manipulation checks, which confirm that a treatment was delivered before its effect is interpreted (Hauser et al., 2018; Ejelöv and Luke, 2020). We apply them to role prompts and then audit the checks themselves.[1]

2 Method

Each of six roles from one council design (Table 1) has three prompt variants: the *operation* prompt, a bare role-name *persona* prompt, and a *neutral* prompt matched in length. Every variant emits the same output schema (atomic claims, counterevidence, failure flags, answer, and confidence), and each role's check is a fixed function of those fields. We ran each role and variant on 200 HaluEval development items (Li et al., 2023) with `gpt-5.6-terra` (primary) and `grok-4.3` (replication), 3,600 calls per family with no exclusions. The preregistered rule required that the paired bootstrap contrast against neutral be significant under Benjamini–Hochberg control across the six roles ($q = 0.05$) *and* reach a paired $d \geq 0.40$.

[1] A preliminary version of the Sec. 3 results is under review at the *Columbia Junior Science Journal*. The audit in Sec. 4 is new.

3 Verdicts

One role passes on `gpt-5.6-terra`. Counter-evidence checking reaches $d = +1.03$, with counterclaims plus flags rising from 0.61 to 1.285 per answer. Two roles pass on `grok-4.3`: counter-evidence ($d = +0.52$) and local-claim checking ($d = +0.69$). On the benchmark side, on 1,000 HaluEval items, three six-agent councils (operation, persona, and neutral prompts) scored within a point of each other (0.839–0.846), and none outperformed a single call (0.861). Every leave-one-out attribution interval contains zero.

4 Auditing the instrument

Guards. Five checks specified a guard against degenerate passing. Two guards were defeatable and three never varied on the observed data; local-claim's was constant on 1,191 of 1,200 records. The only check that passed had no guard.

Discriminant validity (exploratory). A role shows discriminant validity if its prompt moves its own check by at least 0.20 more (in d) than any other check, a criterion fixed before we computed the matrix. This holds in 1 of 12 cells (Table 1). On `grok-4.3`, the counter-evidence prompt moves the explore-then-prune check ($d = -1.23$) more than its own ($+0.52$).

Collapse. Across 3,600 records per family, the local-claim, explore-then-prune, and rare-hypothesis checks correlate at 0.79–1.00 with the number of atomic claims. Six checks are closer to four measures: a prompt that changes how many claims an agent writes moves all three, consistent with the two cross-family sign changes.

Artifact. The explore-then-prune prompt tells the model to record only the hypotheses that *survive* pruning, while its check counts distinct recorded claims as first-pass breadth. The check's $d = -0.95$ (-1.66 on `grok-4.3`) is therefore what the prompt asks for, not evidence that the role explored less. Consistency checking is null on both families but sits near its floor (flag rate 0.025 vs. 0.015), so its null cannot yet be told apart from an insensitive instrument.

Role	Own d vs. neutral terra	Own d vs. neutral grok	Discriminant margin terra	Discriminant margin grok	r with claim count terra	r with claim count grok
Counter-evidence	**+1.03**	**+0.52**	**+0.53**	-0.71	0.17	0.32
Consistency	+0.05	+0.07	-0.22	-1.23	0.03	0.01
Local-claim	-0.18	**+0.69**	-0.75	-0.02	1.00	1.00
Explore-then-prune	-0.95	-1.66	-1.90	-3.30	0.96	0.99
Rare-hypothesis	+0.03	-1.25	-0.74	-2.94	0.79	0.88
Confidence calib.	-0.06	+0.07	-0.34	-0.70	-0.01	-0.01

Table 1 | Bold d: passes the preregistered rule. Margin: own d minus the prompt's largest $|d|$ on another check; bold if ≥ 0.20. Margins and correlations are exploratory (computed on already-observed records).

5 Ongoing work and takeaway

We are preregistering controls that test each check as an instrument. A positive control adds a worked example of the operation, and a negative control instructs the model to avoid it. A check counts as sensitive only if the exemplar exceeds neutral by $d \geq 0.40$ and neutral exceeds the anti-instruction by $d \geq 0.20$. Roles with insensitive checks are reported as uninterpretable, not failed. The plan adds three rewordings per role, a third open-weight family, and an explore-then-prune check that records first-pass hypotheses. Our scope is narrow (six roles, one council design, one benchmark, two families), but the lesson is general: benchmark deltas do not license claims about role mechanisms, and neither does an untested check. A manipulation check should not merely be executable; it should survive one.

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